National Disaster Management Guidelines: Earthquake Management (Part 1)

Issuing Authority: National Disaster Management Authority (NDMA), Government of India | **Document Reference:** NDMA-SOP-EQ-2019-V1

Section 1.1: Scope and Seismic Hazard Profile of India

Over 58 percent of the Indian landmass is vulnerable to moderate to severe seismic ground shaking, spanning Seismic Zones III, IV, and V. High-risk zones include the Himalayan belt, the Andaman & Nicobar islands, the northeastern states, and active seismic fault systems across Kutch and Koyna.

The primary objective of these guidelines is to establish standardized community response measures and architectural resilience standards to reduce fatalities caused by structural collapse, falling facade debris, and secondary post-earthquake utility fires.

Section 1.2: Pre-Earthquake Household Readiness & Disaster Kit

Every household must prepare an emergency Disaster Go-Bag containing essential life-support materials sufficient for at least 72 hours.

- Store at least 3 to 4 liters of clean potable drinking water per person per day in sealed food-grade containers.
- Maintain a comprehensive First Aid kit containing sterile compression bandages, antiseptic solution, medical shears, and a 14-day supply of essential personal prescription medicines.
- Secure heavy domestic furniture, book racks, water heaters, and hanging light chandeliers to structural wall studs using anchor bolts and L-brackets.
- Ensure all family members know the exact physical location of the main domestic LPG cylinder shutoff valve and the primary electrical circuit breaker.

National Disaster Management Guidelines: Earthquake Management (Part 2)

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Section 2.1: Immediate Survival Protocol: Drop, Cover, and Hold On

When ground shaking commences, immediate life-safety action dictates following the DROP, COVER, AND HOLD ON protocol without hesitation.

- DROP immediately to hands and knees to prevent being knocked off your feet by violent ground acceleration.
- COVER your head and neck (and your entire torso if possible) under a sturdy wooden or metal table or heavy desk.
- HOLD ON to your protective shelter with one hand and be prepared to move with it if it shifts across the floor.
- If no sturdy table is nearby, drop along an interior load-bearing partition wall far away from exterior windows, glass panels, mirrors, and tall furniture.
- Keep your face buried against your chest or arm to protect eyes and airways from shattered glass and falling plaster.

Section 2.2: Actions Strictly Prohibited During Active Tremors

Extensive trauma hospital audits demonstrate that the majority of earthquake casualties result from panicked movements during active ground shaking.

- DO NOT attempt to run outside the building while the ground is actively shaking. Falling roof parapets, decorative masonry, air-conditioning units, and facade glass cause catastrophic blunt injuries along building perimeters.
- DO NOT stand in modern interior doorways. Modern doorways are rarely load-bearing and do not provide protection against flying debris.
- DO NOT use passenger or service elevators under any circumstances. Power outages will trap occupants inside elevator shafts.
- DO NOT ignite matches, candles, gas stoves, or cigarette lighters due to the severe threat of cracked domestic gas pipelines.

National Disaster Management Guidelines: Earthquake Management (Part 3)

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Section 3.1: Post-Shaking Evacuation and Secondary Hazard Mitigation

Once tremors halt for at least 60 seconds, occupants must begin methodical post-quake safety procedures.

- Immediately put on sturdy closed-toe shoes or work boots to avoid lacerations from shattered glass and structural nails.
- If a gas leak is smelled or hissing is heard, shut off the main LPG cylinder valve or piped gas meter cock immediately. Open exterior windows to ventilate fumes and evacuate immediately without touching electrical switches.
- Switch off the main electrical breaker ONLY if the distribution board is completely dry and safe to touch.
- Descend multi-story buildings using interior staircases cautiously along the inner railing. Inspect stair landings for diagonal shear cracking.
- Move into an open municipal park, school sports ground, or wide street at least 1.5 times the height of adjacent structures.

Section 3.2: Urban Search & Rescue (USAR) Survival Signaling

If trapped under debris, concrete rubble, or collapsed walls, follow the National Disaster Response Force (NDRF) USAR survival doctrine:

- Cover your nose and mouth with a cloth, handkerchief, or garment to filter lethal pulverized silica concrete dust.
- Tap on metal water pipes, conduits, or solid structural concrete using a stone or piece of rubble in rhythmic bursts of 3 (SOS signal). Sound waves travel effectively through reinforced concrete.
- Use an emergency whistle if accessible. Whistle acoustics carry up to three times further than human vocal cords.
- Avoid continuous shouting. Screaming causes rapid vocal fatigue, panic hyperventilation, and dangerous inhalation of toxic dust particles.